



Modelling Financial Markets

Seminar 19: Review and Project Workshop — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- AI: Artificial Intelligence
- AIC: Akaike Information Criterion
- AR: AutoRegressive (model); in event studies: Abnormal Return
- BET: Bucharest Exchange Trading (index)
- BH: Benjamini–Hochberg (procedure)
- BIC: Bayesian Information Criterion
- BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- CI: Confidence Interval
- COVID-19: COroNaVirus Disease 2019
- DM: Diebold–Mariano (test)
- DQ: Dynamic Quantile (test)
- ES: Expected Shortfall
- ETF: Exchange-Traded Fund
- FDR: False Discovery Rate
- FHS: Filtered Historical Simulation
- FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- FWER: Family-Wise Error Rate
- FZ0: Fissler–Ziegel loss, zero-homogeneous version
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GEO: Government Emergency Ordinance
- GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HS: Historical Simulation
- IGARCH: Integrated GARCH
- IQR: InterQuartile Range

Acronyms used in this material (2/2)

- ▣ MCS: Model Confidence Set
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's Question and Route

- Question of the seminar: **would your project survive the checks reviewed in Chapter 19?** From raw BET closing prices to a backtested VaR 1% with valid inference, and from an idea to a one-page proposal
- Route
 - ▶ First the primer “What You Need for Today” (the lecture comes after the seminar); then A7, A9 on paper: the power of a backtest and multiple testing
 - ▶ B1: the tails and the autocorrelation of the BET; B3: an interval at the GARCH boundary
 - ▶ B5: backtest four VaR 1% models; B7: the look-ahead trap
 - ▶ C2: find the errors in an AI-written backtest; C1, your team's proposal, at home
- What you produce: the A answers, one chart and one table per B exercise, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); the notebook sections carry the same labels as the exercises (A1–A12, B1–B8, C1–C3)
- Seminar: Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

Exercise Map

Exercise	Question	Type	Model
A7	how often does one year catch a bad VaR model?	Solved	–
A9	which strategies survive multiple testing?	Solved	–
B1	how heavy are the BET tails?	Solved	–
B3	which interval for GARCH persistence?	Solved	–
B5	which VaR 1% model would we have trusted?	Solved	–
B7	how much does look-ahead flatter a backtest?	Solved	–
C2	what is wrong in an AI-written backtest?	Proposed	A7, B5, B7
C1	your team's proposal (homework)	Proposed	worked proposal
Extension packs	A1–A6, A8, A10–A12, B2, B4, B6, B8, C3 (optional, after the lecture)	Solved and Proposed (marked on each task)	named on each task

- **[Solved]:** full solution in the slides and in the notebook, a model to copy; **[Proposed]:** you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: verification and inference on real data, each ending with an interpretation question; Part C: the project proposal and AI critique

Data Used

Series	Price used	Calendar	Sample	Used in
BET	official close	Romanian trading days	6 Jan 2000 – 18 Sep 2026, 6 670 returns	A3, B1, B3–B5, B7
Bitcoin	close in USD	every day	18 Sep 2014 – 18 Sep 2026, 4 384 returns	B2, B6, B8
SPY (S&P 500 ETF)	adjusted close	US trading days	2015 – 2026	B8
WIG20	close	Polish trading days	2014 – 2026	C1
Euro Stoxx 50	close	local trading days	2002 – 2018 and later	C3

- Daily prices of the course data, one series per asset (field used: close / adjusted close); duplicate records and weekend-dated rows of the indices removed
- Two series on different calendars (B8, C1): align the *prices* on common trading days first, then compute returns

Setup: Run and Verify

```
bet = 100 * log_returns('bet')
btc = 100 * log_returns('btc')
print(len(bet), bet.index[0].date(),
      bet.index[-1].date())
print(round(bet.std(), 2), len(btc))
```

- The notebook section “Data” defines `log_returns(name)`: daily log returns on the calendar of each series
- Expected output
 - ▶ 6670 2000-01-06 2026-09-18
 - ▶ 1.4 4384
- If your count or first date differs, stop and check the file, the field and the cleaning rules

Tools from Earlier Chapters, Task by Task

- The seminar comes before the lecture; [Chapter 19](#) reviews the course, so almost every tool of today was introduced earlier; only three ideas are new and get a slide each: the power of a backtest, an interval at a boundary, the look-ahead trap
- A9: Holm (FWER) and Benjamini–Hochberg (FDR) corrections, [Chapter 3 Holm \(1979\)](#), [Benjamini & Hochberg \(1995\)](#)
- B1: Hill tail index, robust portmanteau Q^* , moving-block bootstrap, [Chapter 1 Hill \(1975\)](#), [Lobato, Nankervis & Savin \(2001\)](#), [Künsch \(1989\)](#)
- B3: GARCH(1,1) with Student- t innovations, robust standard errors, tests at the boundary $\alpha + \beta = 1$, [Chapter 5 Bollerslev & Wooldridge \(1992\)](#)
- B5, B7: HS, Normal, GARCH- t and FHS VaR; Kupiec, Christoffersen and DQ tests; the FZ0 score, DM tests, the model confidence set, the comparative ES backtest, [Chapter 8 Kupiec \(1995\)](#), [Christoffersen \(1998\)](#), [Engle & Manganelli \(2004\)](#)
 - ▶ FHS (filtered historical simulation): VaR from the GARCH- t $\hat{\sigma}_{t+1}$ times the empirical quantile of the standardised residuals [McNeil & Frey \(2000\)](#)

New Idea 1: Size and Power of a Backtest

- **Size:** probability of rejecting a correct model; **power:** probability of rejecting a wrong one; a backtest is useful only if its power is high for the errors that matter
- With independent breaches, their number is binomial, $X \sim \text{Bin}(T, \pi)$, $P(X = x) = \binom{T}{x} \pi^x (1 - \pi)^{T-x}$; Kupiec rejects $\pi = 1\%$ when $LR_{uc}(x) > 3.84$ [Kupiec \(1995\)](#)
 - ▶ Rejection region $R = \{x \text{ with } LR_{uc}(x) > 3.84\}$; exact size = $\sum_{x \in R} P(X = x \mid \pi = 1\%)$; exact power = $\sum_{x \in R} P(X = x \mid \pi)$ at the true π
 - ▶ X is discrete, so the exact size is not exactly 5%
- Small example, $T = 100$ days (1 breach expected)
 - ▶ $LR_{uc}(0) = -2 \cdot 100 \ln 0.99 = 2.01$, $LR_{uc}(3) = 2.63$, $LR_{uc}(4) = 5.18$: $R = \{x \geq 4\}$, and zero breaches never reject
 - ▶ Size $P(X \geq 4 \mid 1\%) = 0.018$; power 0.14 if the true rate is 2%, 0.74 if it is 5%
- A non-rejection with low power is “no evidence”, not “no problem”; state the power before the test (A7; [Chapter 19](#) after the seminar)

New Ideas 2 and 3: an Interval at a Boundary; the Look-Ahead Trap

- A Wald interval $\hat{p} \pm 1.96 \text{se}(\hat{p})$ can leave the parameter space: GARCH persistence $p = \alpha + \beta \leq 1$, but 0.99 ± 0.02 gives $[0.97; 1.01]$
 - ▶ Near the boundary $\hat{p}$ is not Normal [Andrews \(1999\)](#), and the interval contains impossible values
- **Profile likelihood:** $\ell_P(p) = \max$ of the log-likelihood over the other parameters with p fixed; 95% interval = all p with $2(\ell_{\max} - \ell_P(p)) \leq 3.84$
 - ▶ It stays inside $p \leq 1$; at the boundary point $p = 1$ the statistic is $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$, critical value 2.71 ([Chapter 5](#))
 - ▶ Half-life of a shock $h = \ln 0.5 / \ln p$: 34 days at $p = 0.98$, 69 at $p = 0.99$, 693 at $p = 0.999$, infinite at $p = 1$
- **Look-ahead bias:** a forecast for day t that uses information from after day $t - 1$, e.g. parameters or a quantile estimated on the full sample
 - ▶ Test for your code: every forecast for day t must be computable from a data file truncated at $t - 1$ (B7)

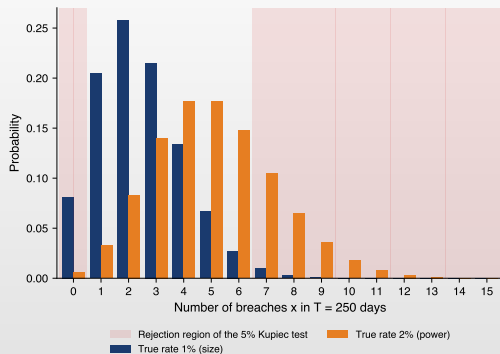
Notation Used in Every Exercise

- ▣ $r_t = 100 \ln(P_t/P_{t-1})$: daily log return in %; loss $L_t = -r_t$
- ▣ $\text{VaR}_\alpha = -q_\alpha(r)$, $\text{ES}_\alpha = -\mathbb{E}[r \mid r \leq q_\alpha(r)]$, positive numbers; the level is the tail probability (VaR 1%, ES 2.5%)
- ▣ Breach (hit) $I_t = \mathbb{1}\{L_t > \text{VaR}_t\}$; with x breaches in T days, Kupiec $LR_{uc} \sim \chi_1^2$ tests the rate [Kupiec \(1995\)](#), Christoffersen $LR_{ind} \sim \chi_1^2$ the independence [Christoffersen \(1998\)](#)
- ▣ Hill estimator on the k largest losses: $\hat{\alpha}^{-1} = \frac{1}{k} \sum_{i=1}^k \ln L_{(i)} - \ln L_{(k+1)}$ [Hill \(1975\)](#); the moment of order p is finite only if $p < \alpha$
- ▣ Robust portmanteau $Q^*(m) = T \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k$, $\hat{\tau}_k = \overline{x_t^2 x_{t-k}^2} / \hat{\gamma}_0^2$, $x_t = r_t - \bar{r}$, $\hat{\gamma}_0 = \overline{x_t^2}$ [Lobato, Nankervis & Savin \(2001\)](#); χ_m^2 calibration needs finite fourth moments
- ▣ Moving-block bootstrap: resample blocks of 20 consecutive days (999 resamples), so that the dependence within blocks is kept [Künsch \(1989\)](#)

A7 [Solved]: The Power of the Kupiec Test — Task

- **Question:** How often does one year of data catch a VaR 1% model whose true breach rate is 2%?
- **Data and assumptions**
 - ▶ A VaR 1% model is backtested over T days; the number of breaches is $X \sim \text{Bin}(T, \pi)$; $H_0: \pi = 0.01$.
 - ▶ Kupiec statistic $LR_{uc}(x) = -2[(T - x) \ln 0.99 + x \ln 0.01 - (T - x) \ln(1 - x/T) - x \ln(x/T)]$ (with $0 \ln 0 = 0$); reject at 5% if $LR_{uc} > \chi^2_{1;0.95} = 3.84$ Kupiec (1995).
- **Tasks**
 - ▶ (a) For $T = 250$, compute $LR_{uc}(x)$ for $x = 0, 1, 2, \dots$ and find the rejection region R ; repeat for $T = 1000$.
 - ▶ (b) Compute the exact size $P(X \in R \mid \pi = 1\%)$ for both T .
 - ▶ (c) Compute the exact power $P(X \in R \mid \pi = 2\%)$ for both T .
- **Report**
 - ▶ The two rejection regions, the exact sizes and the exact powers.
- **Interpretation**
 - ▶ How often does one year of data catch a model whose true breach rate is twice the nominal one, and why does Basel add the traffic-light multiplier?

A7 [Solved]: Solution (1/2)



- (a) Step 1 ($T = 250$, expected 2.5 breaches):
 $LR_{uc}(0) = -2 \cdot 250 \ln 0.99 = 5.03 > 3.84$, so $x = 0$ is rejected; $LR_{uc}(1) = 1.18$.
- Step 2: $LR_{uc}(6) = 3.56 < 3.84$ and $LR_{uc}(7) = 5.50 > 3.84$, so $R = \{0\} \cup \{x \geq 7\}$.
- Step 3 ($T = 1000$, expected 10 breaches): the same scan gives $R = \{x \leq 4\} \cup \{x \geq 17\}$.
- (b) Step 4: at $T = 250$, size
 $= P(X = 0) + P(X \geq 7) = 0.0811 + 0.0137 = 0.095$ under $\pi = 1\%$; at $T = 1000$ the size is 0.055.
- Reading: X is discrete, so the exact size is not 5%; with few expected breaches it can be almost twice the nominal level.

A7 [Solved]: Solution (2/2)

- (c) Step 5: at $T = 250$, power = $P(X = 0) + P(X \geq 7) = 0.0064 + 0.2363 = 0.243$ under $\pi = 2\%$.
- Step 6: at $T = 1\,000$, $P(X \leq 4) + P(X \geq 17) = 0.782$ under $\pi = 2\%$.

	$T = 250$	$T = 1\,000$
Rejection region	$x = 0 \text{ or } x \geq 7$	$x \leq 4 \text{ or } x \geq 17$
Exact size ($\pi = 1\%$)	0.095	0.055
Power ($\pi = 2\%$)	0.243	0.782

- Interpretation: one year of data rejects a model with twice the breach rate only about one time in four; four years are needed for a power near 80%; that is why Basel adds the traffic-light multiplier instead of relying on the test alone.

A9 [Solved]: Holm and Benjamini–Hochberg — Task

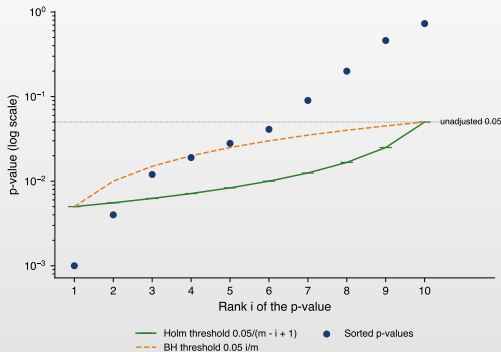
- **Question:** Which of ten trading strategies survive a correction for multiple testing?
- **Data and assumptions**
 - ▶ Ten trading strategies, each tested against a zero excess return; sorted p -values $p_{(1)}, \dots, p_{(10)}$: 0.001, 0.004, 0.012, 0.019, 0.028, 0.041, 0.09, 0.20, 0.46, 0.73; $m = 10$; target level 5%.
 - ▶ FWER: probability of at least one false rejection; $\text{FDR} = \mathbb{E}[V / \max(R, 1)]$, the expected share of false rejections V among all R rejections (0 when nothing is rejected).
- **Tasks**
 - ▶ (a) Count the rejections without correction ($p \leq 0.05$) and with Bonferroni ($p \leq 0.05/m$).
 - ▶ (b) Apply Holm at FWER 5% **Holm (1979)**: compare $p_{(i)}$ with $0.05/(m - i + 1)$ for $i = 1, 2, \dots$ and stop at the first $p_{(i)}$ above its threshold.
 - ▶ (c) Apply Benjamini–Hochberg at FDR 5% **Benjamini & Hochberg (1995)**: find the largest k with $p_{(k)} \leq 0.05k/m$ and reject $H_{(1)}, \dots, H_{(k)}$.
- **Report**
 - ▶ A table with $p_{(i)}$ and both thresholds, and the number of rejections of each procedure.
- **Interpretation**
 - ▶ What does each procedure promise, and which one would you use to screen trading strategies?

A9 [Solved]: Solution (1/2)

i	1	2	3	4	5	6	7	8	9	10
$p_{(i)}$	0.001	0.004	0.012	0.019	0.028	0.041	0.09	0.20	0.46	0.73
Holm	0.0050	0.0056	0.0063	0.0071	0.0083	0.0100	0.0125	0.0167	0.0250	0.0500
BH	0.005	0.010	0.015	0.020	0.025	0.030	0.035	0.040	0.045	0.050

- (a) Unadjusted: 6 p -values below 5%; Bonferroni ($p \leq 0.005$): 2 rejections.
- (b) Holm, step down: $0.001 \leq 0.0050$ and $0.004 \leq 0.0056$ pass; $0.012 > 0.0063$ stops the procedure: 2 rejections.

A9 [Solved]: Solution (2/2)

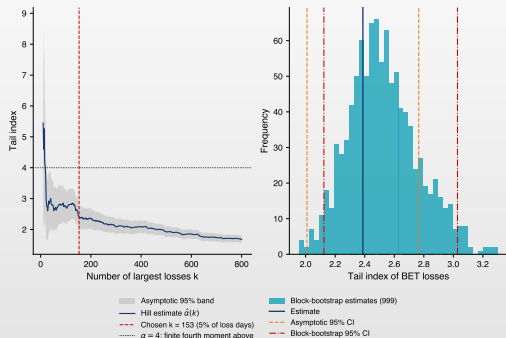


- (c) BH, step up from the largest p : $0.041 > 0.030$, $0.028 > 0.025$, but $0.019 \leq 0.020$, so $k = 4$: 4 rejections (also $0.012 \leq 0.015$ is not needed, since BH rejects all $H_{(i)}$ with $i \leq k$).
- Summary: none 6, Bonferroni 2, Holm 2, BH 4.
- Holm promises: probability of any false discovery $\leq 5\%$ (FWER), uniformly more powerful than Bonferroni.
- BH promises: expected share of false discoveries among the declared ones $\leq 5\%$ (FDR) when the p -values are independent or positively regression dependent on the true nulls [Benjamini & Yekutieli \(2001\)](#); it rejects more.
- Interpretation: for screening many candidate strategies BH is the usual choice (a few false leads are acceptable); before committing capital to one strategy, FWER control (Holm) is the safer promise.

B1 [Solved]: Stylised Facts of BET with Robust Inference — Task

- **Question:** How heavy are the tails of the BET, and does its autocorrelation survive once volatility clustering is accounted for?
- **Data and assumptions**
 - ▶ BET official daily closes (course data), 6 Jan 2000 – 18 Sep 2026: $T = 6\,670$ log returns r_t in %; losses $L_t = -r_t$; moving-block bootstrap, 999 resamples of 20-day blocks; all tests at 5%.
- **Tasks**
 - ▶ (a) Compute the standard deviation, the skewness and the excess kurtosis of r_t .
 - ▶ (b) Estimate the Hill tail index of losses with $k = 5\%$ of the loss days, with the asymptotic 95% CI $\hat{\alpha}(1 \pm 1.96/\sqrt{k})$ and the 95% percentile CI of the block bootstrap.
 - ▶ (c) Compute the Ljung–Box $Q(10)$ on r_t and on r_t^2 and the robust $Q^*(10)$ of the Notation slide; compare each with $\chi_{10,0.95}^2 = 18.31$.
 - ▶ (d) Count the days with $|r_t - \bar{r}| > 4s$ and compare with the number expected under the Normal distribution, $T \cdot 2\Phi(-4)$.
- **Report**
 - ▶ The statistics of (a), $\hat{\alpha}$ with both intervals and a Hill plot, the three portmanteau statistics with p -values and decisions, the count of (d).
- **Interpretation**
 - ▶ Why is the block interval for the tail index wider than the asymptotic one? Does the autocorrelation of r_t survive the robust test, and how far can you trust its p -value given the tail index?

B1 [Solved]: Solution (1/2)



- (a) $s = 1.40\%$, skewness -0.65 , excess kurtosis 11.1 : fat, left-skewed tails.
- (b) Step 1: sort the losses; $k = 153$ (5% of the loss days); $\hat{\alpha} = 2.39$.
- Step 2 (asymptotic): $2.39(1 \pm 1.96/\sqrt{153}) = [2.01; 2.77]$, standard error 0.193 .
- Step 3 (block bootstrap): rebuild 999 series from random 20-day blocks, re-estimate $\hat{\alpha}$ with the same k , take the 2.5% and 97.5% percentiles: $[2.12; 3.03]$, standard error 0.233 .
- Reading: both intervals lie below 4, supporting an infinite fourth moment under the assumed regularly varying tail model, and the sample kurtosis (11.1) is not reported as an estimate (A11).

B1 [Solved]: Solution (2/2)

- (c) $Q(10)$ on r_t : 108.0 ($p < 0.001$) > 18.31 : reject no autocorrelation; robust $Q^*(10) = 24.4$ ($p = 0.007$): still reject, but 4.4 times smaller.
- $Q(10)$ on r_t^2 : 2440 ($p < 0.001$): strong volatility clustering, read descriptively, since if the fourth moment is infinite r_t^2 has no finite variance; first-order autocorrelation of r_t : 0.109 .
- (d) 51 days beyond ± 4 standard deviations, against $6\,670 \cdot 2\Phi(-4) = 0.42$ expected under the Normal distribution.
- Interpretation 1: extremes cluster in crises; blocks keep the clusters, so the block interval accounts for the dependence that the asymptotic formula ignores.
- Interpretation 2: the robust test still rejects, but with $\hat{\alpha} < 4$ its χ^2 calibration is not guaranteed: sample autocorrelations of heavy-tailed GARCH-type returns converge more slowly and to non-Normal limits [Davis & Mikosch \(1998\)](#); read it as evidence of a small autocorrelation; stale prices of less liquid stocks ([Chapters 2, 10](#)) are a plausible explanation that needs its own test.

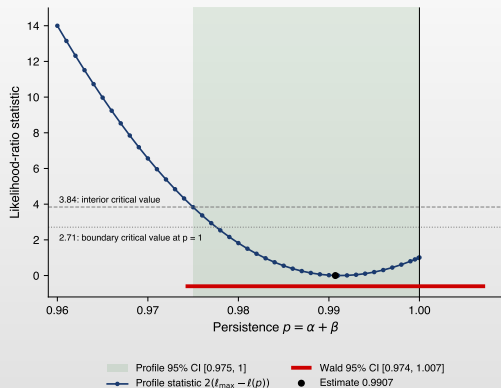
B3 [Solved]: GARCH(1,1)- t for BET: an Interval at the Boundary — Task

- **Question:** Which interval for GARCH persistence can be reported when the estimate is close to 1?
- **Data and assumptions**
 - ▶ BET daily log returns in % (B1); constant mean; $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$ with standardised Student- t innovations; robust standard errors [Bollerslev & Wooldridge \(1992\)](#); level 95%.
 - ▶ Persistence $p = \alpha + \beta$: a finite unconditional variance needs $p < 1$ (strict stationarity only $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ [Nelson \(1990\)](#)); $\hat{p}$ lies near the boundary $p = 1$ of the parameter space $p \leq 1$ [Andrews \(1999\)](#).
- **Tasks**
 - ▶ (a) Estimate the model by maximum likelihood; report $\hat{\alpha}$, $\hat{\beta}$, $\hat{\nu}$ with robust standard errors and $\hat{p}$.
 - ▶ (b) Compute $\text{se}(\hat{p})$ from the robust covariance of $(\hat{\alpha}, \hat{\beta})$ and the Wald CI $\hat{p} \pm 1.96 \text{se}(\hat{p})$.
 - ▶ (c) Profile likelihood: for each p in 0.960, 0.961, ..., 0.999, 0.9995, 0.9999, 1 maximise the log-likelihood over μ , ω , ν and $s = \alpha/p$; keep $p < 1$ if $2(\ell_{\max} - \ell(p)) \leq 3.84$, and $p = 1$ if the statistic is ≤ 2.71 , the 5% critical value of $\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ [Andrews \(1999\)](#).
 - ▶ (d) Transform the estimate and both intervals into half-lives $\ln 0.5 / \ln p$; (e) compare $t_{\hat{\nu}}^{-1}(0.01)\sqrt{(\hat{\nu} - 2)/\hat{\nu}}$ with $z_{0.01}$.
- **Report**
 - ▶ Estimates with standard errors; both CIs for p and for the half-life; the profile curve; the two quantiles.
- **Interpretation**
 - ▶ Which interval can you report, and why is the other one wrong? What does $\hat{\nu} \approx 5$ mean for VaR?

B3 [Solved]: Solution (1/2)

- (a) $\hat{\alpha} = 0.192$ (se 0.025), $\hat{\beta} = 0.799$ (se 0.024), $\hat{\nu} = 5.01$ (se 0.31); $\hat{p} = 0.9907$.
- (b) With the robust covariance, $\text{se}(\hat{p}) = 0.0084$ (the negative covariance of $\hat{\alpha}$ and $\hat{\beta}$ makes it smaller than either standard error).
- Wald CI: $0.9907 \pm 1.96 \cdot 0.0084 = [0.974; 1.007]$: the upper end is above 1, outside the parameter space $p \leq 1$.
- Why it is wrong: at the boundary the distribution of $\hat{p}$ is not Normal [Andrews \(1999\)](#), and the interval contains values $p > 1$, for which the variance explodes and the half-life is not defined.
- (d) Half-life $\ln 0.5 / \ln 0.9907 = 74$ days; from the Wald interval: 26 days at the lower end and no value at the upper end.

B3 [Solved]: Solution (2/2)



- (c) Profile statistic $2(\ell_{\max} - \ell(p))$: 4.32 at $p = 0.974$ (excluded), 3.83 at $p = 0.975$ (included), 0.99 at $p = 0.9999$ (included); at $p = 1$: $1.01 < 2.71$, boundary p -value 0.16 (included).
- Profile CI [0.975; 1]: the likelihood excludes no value from 0.975 up to and including IGARCH, $p = 1$.
- (d) Half-life interval: at least $\ln 0.5 / \ln 0.975 = 27$ days, with no finite upper bound, since $p = 1$ is in the interval.
- (e) $t_{\hat{\nu}}^{-1}(0.01) \sqrt{(\hat{\nu} - 2)/\hat{\nu}} = -2.606$ against $z_{0.01} = -2.326$: for the same σ_t the Student- t VaR 1% is 1.12 times the Normal one (zero mean).
- Interpretation 1: report the profile interval; shocks last at least a month, but the data cannot tell a month from several years, and IGARCH ($p = 1$) is not rejected.
- Interpretation 2: $\hat{\nu} \approx 5$ means heavy tails even conditionally on σ_t ; a Normal GARCH VaR 1% would be too low.

B5 [Solved]: Backtesting VaR 1% and ES 2.5% for BET, 2005–2026 — Task

- **Question:** Which of four VaR 1% models for the BET would we have trusted in real time, 2005–2026?
- **Data and assumptions**
 - ▶ BET daily losses $L_t = -r_t$, evaluation 3 Jan 2005 – 18 Sep 2026, $T = 5\,432$ days; each forecast uses only data up to $t - 1$.
 - ▶ Four one-day models [McNeil & Frey \(2000\)](#): HS (500 days), Normal (500 days), GARCH- t (1000-day window, re-estimated every 21 days), FHS (GARCH- t filter, empirical quantile of the standardised residuals); VaR 1% and the pair (VaR 2.5%, ES 2.5%).
- **Tasks**
 - ▶ (a) Count the breaches $L_t > \text{VaR}_t$; compute the rate, the Kupiec p_{uc} and the Christoffersen p_{ind} [Kupiec \(1995\)](#), [Christoffersen \(1998\)](#).
 - ▶ (b) Run the DQ test [Engle & Manganelli \(2004\)](#): regress $I_t - 0.01$ on a constant, 4 lagged hits and VaR_t ; $DQ \sim \chi^2(6)$.
 - ▶ (c) Compute the mean FZ0 score of (VaR 2.5%, ES 2.5%) [Patton, Ziegel & Chen \(2019\)](#), Diebold–Mariano tests (Newey–West, 9 lags) and the 10% model confidence set [Hansen, Lunde & Nason \(2011\)](#).
 - ▶ (d) Comparative ES backtest against HS [Nolde & Ziegel \(2017\)](#): with t the DM statistic of $\text{FZ0}(\text{model}) - \text{FZ0}(\text{HS})$, red if $t > 1.645$, green if $t < -1.645$, yellow otherwise.
- **Report**
 - ▶ A coverage table for (a)–(b), a chart of cumulative breaches, a table of scores, MCS p -values, ES zones and mean VaR.
- **Interpretation**
 - ▶ Which model meets all the backtesting criteria of this exercise, and which one would a trader prefer? Do the formal comparisons confirm the backtests?

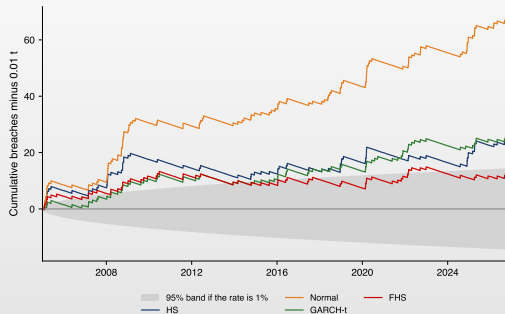
B5 [Solved]: Solution (1/3)

- (a)–(b) Steps: $\text{Hit}_t = \mathbb{1}\{L_t > \text{VaR}_t\}$; expected breaches $0.01 \cdot 5432 \approx 54$; tests at 5%.

Model	x	Rate (%)	p_{uc}	p_{ind}	p_{DQ}
HS	79	1.45	0.002	<0.001	<0.001
Normal	122	2.25	<0.001	<0.001	<0.001
GARCH- t	80	1.47	0.001	0.864	0.007
FHS	67	1.23	0.095	0.267	0.129

- HS and Normal fail coverage, independence and DQ: they react too slowly and their breaches cluster.
- For GARCH- t independence is not rejected, but it has too many breaches, so it fails coverage and DQ; only FHS is rejected by none of the three tests.

B5 [Solved]: Solution (2/3)



- Chart: cumulative breaches minus the expected $0.01 t$; a line above the grey band has too many breaches for a 1% model
- At the end: Normal 122, HS 79, GARCH- t 80, FHS 67 breaches, against about 54 expected; only FHS ends inside the band
- The Normal line jumps in 2008, 2020 and 2022: its breaches arrive in clusters, the pattern that the independence and DQ tests detect

B5 [Solved]: Solution (3/3)

- (c)–(d) Lower FZ0 is better; DM t vs HS < -1.645 means the model beats HS.

Model	FZ0	p_{MCS}	DM t vs HS	ES zone	Mean VaR (%)
HS	1.424	0.00	–	standard	3.66
Normal	1.495	0.00	2.62	red	2.90
GARCH- t	1.175	0.18	–5.75	green	2.95
FHS	1.157	1.00	–5.92	green	3.17

- No statistically detectable FZ0 difference between FHS and GARCH- t (DM $t = 1.39$); together they form the model confidence set {GARCH- t , FHS}; Normal is red, GARCH- t and FHS are green.
- Interpretation 1: FHS is the only model that meets all the stated criteria (coverage, independence, DQ, green ES zone), which summarises the evaluation criteria used in this exercise (not the regulatory checklist of the [Basel backtesting standard](#)); a trader wants low capital, and the Normal model has the lowest average VaR (2.90% vs FHS 3.17%) while failing every test.
- Interpretation 2: the formal comparisons confirm the backtests and add one thing: GARCH- t fails coverage, but its FZ0 score for (VaR, ES) is not significantly different from that of FHS.

B7 [Solved]: The Look-Ahead Trap — Task

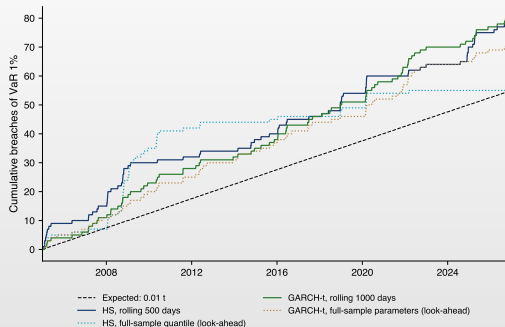
- **Question:** How much better does a backtest look when it uses information from the future?
- **Data and assumptions**
 - ▶ BET losses of B5: evaluation 3 Jan 2005 – 18 Sep 2026, $T = 5432$ days, $0.01T \approx 54$ expected breaches; rolling HS and GARCH- t results from B5.
 - ▶ The look-ahead versions use data that were not available at $t - 1$; Kupiec test at 5%.
- **Tasks**
 - ▶ (a) HS with look-ahead: set every day $\text{VaR}_{1\%} = q_{0.99}(L)$, the empirical quantile of all losses of the evaluation period; count the breaches and compute the rate.
 - ▶ (b) GARCH- t with look-ahead: estimate once on the full sample 6 Jan 2000 – 18 Sep 2026, filter $\hat{\sigma}_t$ day by day, set $\text{VaR}_t = -(\hat{\mu} + \hat{\sigma}_t \tilde{t}_\nu^{-1}(0.01))$ with $\tilde{t}$ the standardised Student- t ; count the breaches and run the Kupiec test.
 - ▶ (c) Compare both with the rolling versions of B5 in one table.
- **Report**
 - ▶ A table with breaches, rate and p_{uc} for the four versions.
- **Interpretation**
 - ▶ Why does the look-ahead version look better? Could anyone have used it in 2008?

B7 [Solved]: Solution (1/2)

- (a) Full-sample HS: $\text{VaR}_{1\%} = 4.24\%$ every day; 55 breaches, rate 1.01%: about 1% by construction, because 1% of the losses of a sample exceed its own 0.99 quantile.
- (b) Full-sample GARCH- t : 71 breaches, rate 1.31%, p_{uc} 0.030: still rejected at 5%, but much closer to 1% than the rolling model.

Model	Parameters	x	Rate (%)	p_{uc}
HS	full sample (look-ahead)	55	1.01	0.93
HS	rolling, 500 days	79	1.45	0.002
GARCH- t	full sample (look-ahead)	71	1.31	0.030
GARCH- t	rolling, 1000 days	80	1.47	0.001

B7 [Solved]: Solution (2/2)



- (c) Reading: both look-ahead versions have fewer breaches than their rolling counterparts and look better calibrated.
- Why: the full sample includes 2008, 2020 and every later crash, so the quantile and the parameters have already seen the losses they are tested on.
- Interpretation: no one could have used these numbers in 2008, since they depend on losses from 2008 to 2026; only the rolling result is evidence about a forecasting model.
- Check for your own code: every forecast for day t must be computable from a data file truncated at $t - 1$.

C2 [Proposed]: Critique an AI-Written Backtest — the Answer

Solution discussed in class

- ▣ **Student prompt:** “bet holds the daily BET closes. Write a backtest of historical-simulation VaR 1% with a 500-day window and a Kupiec test, and give me a reference for the result.”
- ▣ **Answer of the AI assistant**

```
import numpy as np
from scipy.stats import chi2
r = np.log(bet).diff().dropna()
var = -r.rolling(500).quantile(0.01)
hit = (-r > var)[var.notna()]
x, n, p = hit.sum(), hit.size, 0.01
pi = x / n
lr = -2 * (x*np.log(p) + (n-x)*np.log(1-p) - x*np.log(pi) - (n-x)*np.log(1-pi))
print(x, n, chi2.sf(lr, df=2))
```

- ▣ “The breach rate is close to 1% and the Kupiec p -value is above 5%: historical simulation is unbiased for BET, as shown by Kupiec and Christoffersen (2004), *Backtesting historical simulation VaR*, Journal of Risk Model Validation, 8(2), 45–67.”

C2 [Proposed]: Critique an AI-Written Backtest — Task

Solution discussed in class

- ▣ The answer on the previous slide contains at least **three** errors that often appear in AI-written project code; also assess its conclusion and check how the code behaves when there are zero breaches
 - ▶ Model: B5 and B7 (backtest and the look-ahead trap), A7 (Kupiec test)
- ▣ Your tasks
 - ▶ (a) Find each error: which line or sentence, and why it is wrong
 - ▶ (b) Say how you would detect it without knowing the answer: a timing test, the number of restrictions, a DOI search
 - ▶ (c) Give the corrected code and a correct reference
 - ▶ (d) Log each error as in [AI_ERRORS.md](#): what the AI answered, what was wrong, how you found out, the correct version

Proposal Template (One Page)

Sections

- **Question:** one sentence; H_0 , H_1 and the benchmark to beat
- **Replication target:** the table or figure you will reproduce, with DOI, page and table number
- **Data:** series, owner, period, frequency, cleaning rules
- **Identification:** control market, placebo dates, break test with unknown date, or out-of-sample design

Sections (continued)

- **Tests and power:** the main test, its loss, and the minimum detectable effect at your T
- **Specification grid:** every window, lag, α and sample you will try, and the multiplicity correction
- **Risks:** data problems, too few events, overfitting
- **References:** 10–15, at least 3 from the last five years, each checked to exist (DOI or link)

A Worked Proposal (1/2)

- **Question:** did the tail risk of the BET change after the FTSE reclassification of the BVB as a Secondary Emerging market (21 September 2020)?
- **Hypothesis:** H_0 : same Hill tail index of losses and same VaR 1% breach rates in the six years before and after
- **Data:** BET official closes, 22 Sep 2014 – 18 Sep 2026; control: WIG20 (Poland); duplicate records (rows identical to the previous one) and weekend-dated rows removed; BET and WIG20 prices aligned on common trading days first, then returns
- **Identification:** difference-in-differences of Hill indices (BET minus WIG20); placebo dates $\pm$ one year [Andrews \(1993\)](#)
 - ▶ Identifying assumptions: without the event the two tail indices would have changed alike; no spillover of the reclassification to Poland; no concurrent shock specific to one of the two markets
 - ▶ Complementary scale diagnostic: a sup-Wald break test with unknown date on $\ln |r_t|$ (it detects shifts in volatility, not in the tail index)

A Worked Proposal (2/2)

- **Inference:** moving-block bootstrap (999 resamples, 20-day blocks)
 - ▶ Within each period the same blocks of days for BET and WIG20, so the dependence between the two markets is kept; independent draws in the two disjoint periods
- **Power:** minimum detectable difference-in-differences at 80% power and 5% level, $(z_{0.975} + z_{0.80}) se = 2.80 se$ (the standard 80%-power formula; C1(e))
- **Specification grid:** $k = 2.5\%, 5\%, 10\%$ of the loss days $\times$ blocks of 10, 20, 40 days: 9 tests of the difference-in-differences, Holm correction (A9)
- **Replication target:** not part of this illustration, which is a partial model; your proposal must name the published table or figure, with DOI and location (C1(b))
- **Risks:** the pre-period contains the COVID-19 crash and the BET fall of 19 December 2018, after the announcement of the bank-asset tax (GEO 114/2018); one event, so no causal claim without the control

C1 [Proposed]: Your Team's Proposal — Task (1/2)

Solution discussed in class

- ▣ **Question:** What will your team study, and can the design answer the question?
- ▣ **Data and assumptions**
 - ▶ One market or asset of your choice: a series from the course data (the list is in the course repository) or a public source named by its official owner; daily data unless the question needs another frequency.
 - ▶ Use the template of the previous slides; the worked proposal is a partial model (it leaves out the replication target); ideas: Part C of Seminars 0–18.
- ▣ **Tasks**
 - ▶ (a) Write the question in one sentence, with H_0 , H_1 and the benchmark to beat.
 - ▶ (b) Name the published table or figure you will replicate first, with DOI, page and table number.
 - ▶ (c) Specify the data: series, owner, period, frequency and cleaning rules.
 - ▶ (d) Describe the identification: control market, placebo dates, break test with unknown date, or out-of-sample design.
 - ▶ (e) State the main test and its loss function, and compute the smallest effect your design detects with 80% power at your T (as in A7).
 - ▶ (f) List the specification grid (windows, lags, α levels, samples), count the tests and choose the multiplicity correction (A9–A10).

C1 [Proposed]: Your Team's Proposal — Task (2/2)

Solution discussed in class

□ Report

- ▶ The filled template on one page, with 10–15 checked references, within one week.
- ▶ Model: the worked proposal

□ Checklist for the proposal

- ▶ Which published table or figure will you replicate first (step 1 of the project)?
- ▶ Is the question answerable with data you already have, and is the benchmark strong?
- ▶ Is the loss function fixed in advance, and does every forecast use only information available at $t - 1$?
- ▶ Have you counted how many models and tests you will run, and chosen the correction?

Team Project: Proposal and Repository

- C1: the one-page proposal, within one week
- README: question, data, how to run
- The pre-analysis plan as a commit, before the test period is opened
- `AI_USE.md` and `AI_ERRORS.md`: every AI use and every caught error, as in C2
- Extension packs are optional; further study:  [Statistics of Financial Markets](#)

Extension Packs: What Each One Needs

Pack	Exercises	Prerequisites
1. Volatility inference	A1 [Solved], A2, A3 [Solved], A4, B4	B3
2. Risk measures and backtests	A5 [Solved], A6, A8, B6	A7, B5
3. Tails and multiple testing	A10, A11 [Solved], B2, A12	A9, B1
4. Data traps	B8	B7
5. Nonstandard errors	C3	A9, the lecture case study

- Do the exercises of a pack in the order shown; each one names its model, and the notebook sections carry the same labels
- Useful for the team project: C3 is a possible project start

A1 [Solved]: The QMLE Sandwich for GARCH(1,1) — Task

- **Question:** How much do Hessian-only standard errors understate GARCH uncertainty when the innovations have heavy tails?
- **Data and assumptions**
 - ▶ Model: $r_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$, $\theta = (\omega, \alpha, \beta)^\top$; z_t i.i.d. with $\mathbb{E}z_t = 0$, $\mathbb{E}z_t^2 = 1$, $\kappa = \mathbb{E}z_t^4 < \infty$.
 - ▶ Estimation by the Gaussian quasi-likelihood $\ell_t(\theta) = -\frac{1}{2}(\ln \sigma_t^2 + r_t^2/\sigma_t^2)$; notation $g_t = \sigma_t^{-2} \partial \sigma_t^2 / \partial \theta$, known at $t-1$ [Bollerslev & Wooldridge \(1992\)](#).
 - ▶ For (c): the GARCH(1,1)- t fit to BET in B3 gives $\hat{\nu} = 5.011$; for Student- t innovations $\kappa = 3(\nu - 2)/(\nu - 4)$ when $\nu > 4$.
- **Tasks**
 - ▶ (a) Differentiate ℓ_t with respect to θ , write the score $s_t = \partial \ell_t / \partial \theta$ in terms of z_t and g_t , and show that $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] = 0$.
 - ▶ (b) Compute $A = -\mathbb{E}[\partial s_t / \partial \theta^\top]$ and $B = \mathbb{E}[s_t s_t^\top]$ using $\mathbb{E}z_t^2 = 1$ and $\mathbb{E}z_t^4 = \kappa$, and show that the sandwich equals $A^{-1} B A^{-1} = \frac{\kappa-1}{2} A^{-1}$.
 - ▶ (c) Substitute $\hat{\nu} = 5.011$: compute κ , the correction factor $(\kappa - 1)/2$ for variances and $\sqrt{(\kappa - 1)/2}$ for standard errors.
- **Report**
 - ▶ The expressions for s_t , A and B ; the values of κ and of both correction factors.
- **Interpretation**
 - ▶ When are the Hessian-only standard errors (square roots of the diagonal of A^{-1}/T) correct, by how much do they inflate the t -statistics for BET, and what happens when $\nu \leq 4$?

A1 [Solved]: Solution (1/2)

- (a) Step 1: $\partial \ell_t / \partial \sigma_t^2 = -\frac{1}{2} \left(\frac{1}{\sigma_t^2} - \frac{r_t^2}{\sigma_t^4} \right) = \frac{1}{2\sigma_t^2} (z_t^2 - 1)$.
- Step 2 (chain rule): $s_t = \frac{1}{2}(z_t^2 - 1) g_t$; g_t is known at $t - 1$ and $\mathbb{E}[z_t^2 - 1] = 0$, so $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] = 0$: the score is a martingale difference.
- (b) Step 3: $\partial z_t^2 / \partial \theta = -z_t^2 g_t$, so $\frac{\partial s_t}{\partial \theta^\top} = \frac{1}{2}(z_t^2 - 1) \frac{\partial g_t}{\partial \theta^\top} - \frac{1}{2} z_t^2 g_t g_t^\top$.
- Step 4: the first term has conditional mean 0 and $\mathbb{E} z_t^2 = 1$, hence $A = \frac{1}{2} \mathbb{E}[g_t g_t^\top]$.
- Step 5: $\mathbb{E}(z_t^2 - 1)^2 = \kappa - 2 + 1 = \kappa - 1$, hence $B = \frac{1}{4}(\kappa - 1) \mathbb{E}[g_t g_t^\top] = \frac{\kappa - 1}{2} A$.

A1 [Solved]: Solution (2/2)

- Step 6: $A^{-1}BA^{-1} = \frac{\kappa-1}{2}A^{-1}AA^{-1} = \frac{\kappa-1}{2}A^{-1}$; it equals the Hessian-only matrix A^{-1} only if $\kappa = 3$, the kurtosis of the Normal distribution.
- (c) Step 7: $\kappa = 3(\hat{\nu} - 2)/(\hat{\nu} - 4) = 3 \cdot 3.011/1.011 = 8.93$.
- Step 8: variances $\times (\kappa - 1)/2 = 3.97$; standard errors $\times \sqrt{3.97} = 1.99$.
- Interpretation: with BET-like tails the Hessian-only standard errors are about half the correct ones, so their t -statistics are about twice too large; report the sandwich [Bollerslev & Wooldridge \(1992\)](#).
- With $\nu \leq 4$, $\kappa = \infty$, B does not exist and the QMLE is no longer $\sqrt{T}$ -Normal: the usual intervals lose their justification.

A2 [Proposed]: The Delta Method for the Half-Life — Task

Solution discussed in class

- ▣ **Question:** Can a delta-method interval describe the uncertainty of a volatility half-life?
- ▣ **Data and assumptions**
 - ▶ Half-life of a volatility shock $h(p) = \ln 0.5 / \ln p$, where $p = \alpha + \beta$ is the persistence of a GARCH(1,1).
 - ▶ BET estimates of B3 with robust covariance: $\hat{p} = 0.9907$, $\text{se}(\hat{p}) = 0.0084$; $\hat{p}$ is asymptotically Normal (A1); level 95%.
- ▣ **Tasks**
 - ▶ (a) Derive $h'(p)$ and write the delta-method standard error $\text{se}(\hat{h}) = |h'(\hat{p})| \text{se}(\hat{p})$.
 - ▶ (b) Compute $\hat{h}$, $|h'(\hat{p})|$, $\text{se}(\hat{h})$ and the Wald interval $\hat{h} \pm 1.96 \text{se}(\hat{h})$.
 - ▶ (c) Using $\ln p \approx -(1-p)$, show that $|h'(p)| \approx \ln 2 / (1-p)^2$ as $p \rightarrow 1$; evaluate the exact and the approximate slope at $p = 0.99, 0.995$ and 0.999 .
- ▣ **Report**
 - ▶ $\hat{h}$ with its standard error and Wald interval; a table of $|h'(p)|$, exact and approximate.
- ▣ **Interpretation**
 - ▶ Why is a Normal interval for the half-life meaningless here, and what should you report instead?
- ▣ **Model:** A1

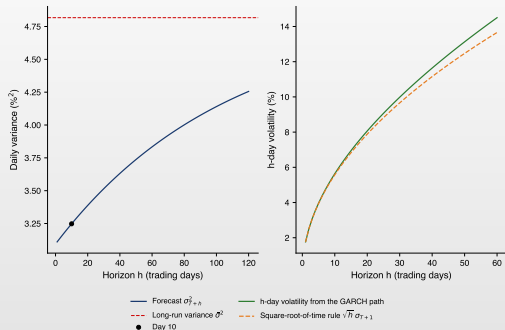
A3 [Solved]: GARCH(1,1) Forecast for BET — Task

- **Question:** How much risk does the BET carry over the next 10 days according to GARCH, and does the square-root-of-time rule get it right?
- **Data and assumptions**
 - ▶ BET daily log returns in % (official closes, course data), 6 Jan 2000 – 18 Sep 2026; GARCH(1,1)- t on the full sample: $\omega = 0.0448$, $\alpha = 0.1916$, $\beta = 0.7991$.
 - ▶ Tomorrow's variance, the one-step forecast $\sigma_{T+1}^2 = \omega + \alpha \varepsilon_T^2 + \beta \sigma_T^2 = 3.11$ ($\sigma_{T+1} = 1.76\%$); forecasts $\sigma_{T+h}^2 = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{T+1}^2 - \bar{\sigma}^2)$; 252 trading days a year.
- **Tasks**
 - ▶ (a) Compute the persistence $\alpha + \beta$, the long-run variance $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$, the daily volatility $\bar{\sigma}$ and the annualised volatility $\sqrt{252} \bar{\sigma}$.
 - ▶ (b) Compute the half-life of a shock $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$.
 - ▶ (c) Forecast the variance on day 10, σ_{T+10}^2 .
 - ▶ (d) Compute the variance of the 10-day return $\sum_{h=1}^{10} \sigma_{T+h}^2$ with the closed form of the geometric sum, the 10-day volatility, and compare it with the square-root-of-time rule $\sqrt{10} \sigma_{T+1}$.
- **Report**
 - ▶ Every number of (a)–(d), each with the substituted formula.
- **Interpretation**
 - ▶ Why does the forecast move towards the long-run level, and when does the square-root-of-time rule understate or overstate the 10-day risk?

A3 [Solved]: Solution (1/2)

- (a) Step 1: $\alpha + \beta = 0.9907$, so $1 - \alpha - \beta = 0.0093$.
- Step 2: $\bar{\sigma}^2 = 0.0448/0.0093 = 4.82$, i.e. $\bar{\sigma} = 2.19\%$ a day.
- Step 3: $\sqrt{252 \cdot 4.82} = \sqrt{1214.6} = 34.9\%$ a year.
- (b) Step 4: $h_{1/2} = \ln 0.5 / \ln 0.9907 = 74.2$ trading days, about 3.5 months of 21 trading days.
- Reading: a shock to variance loses half of its effect in about 74 trading days, so BET volatility is very persistent.

A3 [Solved]: Solution (2/2)



- (c) Step 5: gap to the long run
 $\sigma^2_{T+1} - \bar{\sigma}^2 = 3.11 - 4.82 = -1.71$; $(\alpha + \beta)^9 = 0.919$.
- Step 6: $\sigma^2_{T+10} = 4.82 + 0.919 \cdot (-1.71) = 3.25$: today is below the long run, so the forecast rises.
- (d) Step 7: $\sum_{h=1}^{10} \sigma^2_{T+h} = 10\bar{\sigma}^2 + (\sigma^2_{T+1} - \bar{\sigma}^2) \frac{1 - (\alpha + \beta)^{10}}{1 - \alpha - \beta} = 10 \cdot 4.82 + (-1.71) \cdot 9.592 = 31.80$.
- Step 8: 10-day volatility $\sqrt{31.80} = 5.64\%$, against $\sqrt{10} \cdot 1.76 = 5.57\%$ by the square-root-of-time rule.
- Interpretation: variance reverts to $\bar{\sigma}^2$ at rate $\alpha + \beta$; the square-root-of-time rule understates the 10-day risk when volatility is below its long-run level (as now) and overstates it after a crisis spike.

A4 [Proposed]: GARCH Forecast with Given Parameters — Task

Solution discussed in class

- ▣ **Question:** The same forecast when tomorrow's volatility is above its long-run level
- ▣ **Data and assumptions**
 - ▶ A GARCH(1,1) for daily returns in % with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$; tomorrow's variance $\sigma_{T+1}^2 = 4$ ($\sigma_{T+1} = 2\%$).
 - ▶ Forecast formula and geometric sum as in A3; 252 trading days a year.
- ▣ **Tasks**
 - ▶ (a) Compute the persistence, the long-run variance $\bar{\sigma}^2$ and the annualised long-run volatility.
 - ▶ (b) Compute the half-life of a shock.
 - ▶ (c) Forecast the variance on day 5, σ_{T+5}^2 .
 - ▶ (d) Compute the variance of the 5-day return and the 5-day volatility, and compare them with $5\sigma_{T+1}^2 = 20$ and $\sqrt{5}\sigma_{T+1}$.
- ▣ **Report**
 - ▶ Every number of (a)–(d) with the substituted formula.
- ▣ **Interpretation**
 - ▶ Is the 5-day volatility above or below $\sqrt{5}\sigma_{T+1}$, and why?
- ▣ **Model:** A3

B4 [Proposed]: Asymmetry: GJR-GARCH- t for BET — Task

Solution discussed in class

- ▣ **Question:** Do negative shocks raise BET volatility more than positive ones?
- ▣ **Data and assumptions**
 - ▶ BET returns as in B3; GJR-GARCH(1,1)- t : $\sigma_t^2 = \omega + (\alpha + \gamma \mathbb{I}\{\varepsilon_{t-1} < 0\})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$; the GARCH(1,1)- t of B3 is the restricted model $\gamma = 0$.
 - ▶ $T = 6\,670$; $\text{AIC} = -2\ell + 2k$, $\text{BIC} = -2\ell + k \ln T$ (k = number of parameters); level 5%.
- ▣ **Tasks**
 - ▶ (a) Estimate the GJR model and report $\hat{\gamma}$, its robust standard error and $t = \hat{\gamma}/\text{se}(\hat{\gamma})$.
 - ▶ (b) Compute $LR = 2(\ell_{\text{GJR}} - \ell_{\text{GARCH}})$ and its p -value from $\chi^2(1)$.
 - ▶ (c) Compute AIC and BIC of both models and show that $\Delta\text{AIC} = 2 - LR$ and $\Delta\text{BIC} = \ln T - LR$ (GJR minus GARCH).
 - ▶ (d) Compute $(\hat{\alpha} + \hat{\gamma})/\hat{\alpha}$, the impact of a negative shock relative to a positive one.
- ▣ **Report**
 - ▶ $\hat{\gamma}$ with standard error and t ; LR and p ; both criteria for both models; the ratio of (d).
- ▣ **Interpretation**
 - ▶ Do the tests and the criteria agree? Which model would you use for VaR, and how would you decide?
- ▣ **Model:** B3

A5 [Solved]: Normal VaR 1% and ES 2.5% — Task

- **Question:** How large are VaR 1% and ES 2.5% of a BET position under the Normal distribution?
- **Data and assumptions**
 - ▶ A position of 1 million RON in the BET; one-day log return $r \sim N(\mu, \sigma^2)$ with $\mu = 0.05\%$, $\sigma = 1.4\%$; loss $L = -r$; amounts in RON use the linear approximation $10^6 \cdot L/100$ (exact loss: $10^6 [1 - e^{r/100}]$).
 - ▶ Definitions: $\text{VaR}_\alpha = -q_\alpha(r)$, $\text{ES}_\alpha = -\mathbb{E}[r \mid r \leq q_\alpha(r)]$; for the Normal distribution $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ and $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$, with φ the standard Normal density.
- **Tasks**
 - ▶ (a) Find $z_{0.01}$ and compute $\text{VaR}_{1\%}$ in % and in RON.
 - ▶ (b) Find $z_{0.025}$ and $\varphi(z_{0.025})$ and compute $\text{ES}_{2.5\%}$ in % and in RON.
 - ▶ (c) Compute the ratio $\text{ES}_{2.5\%}/\text{VaR}_{1\%}$.
- **Report**
 - ▶ Both measures in % and in RON, and their ratio.
- **Interpretation**
 - ▶ Why did the Basel Committee replace VaR 1% by ES 2.5%, if the two numbers are almost equal under the Normal distribution?

A5 [Solved]: Solution

- (a) Step 1: $z_{0.01} = -2.326$, so $\sigma z_{0.01} = 1.4 \cdot (-2.326) = -3.257$.
- Step 2: $\text{VaR}_{1\%} = -(0.05 - 3.257) = 3.207\%$, i.e. $3.207\% \cdot 10^6 = 32\,069$ RON.
- (b) Step 3: $z_{0.025} = -1.960$, $\varphi(z_{0.025}) = 0.0584$, $\varphi(z_{0.025})/0.025 = 2.338$.
- Step 4: $\text{ES}_{2.5\%} = -0.05 + 1.4 \cdot 2.338 = -0.05 + 3.273 = 3.223\%$, i.e. $32\,229$ RON.
- (c) Step 5: $\text{ES}_{2.5\%}/\text{VaR}_{1\%} = 3.223/3.207 = 1.005$: under the Normal distribution the two numbers are almost equal.
- Interpretation: the switch is not about the level under the Normal distribution; ES looks inside the tail and is subadditive, while VaR only marks the entrance of the tail ([Chapter 7](#)); with heavy tails ES rises more (A6).

A6 [Proposed]: Student- t VaR 1% and ES 2.5% — Task

Solution discussed in class

- ▣ **Question:** How do the two measures change when the tails are heavy but the variance is the same?
- ▣ **Data and assumptions**
 - ▶ Same position, $\mu = 0.05\%$ and $\sigma = 1.4\%$ as in A5, but $r = \mu + s T_\nu$, T_ν Student- t with $\nu = 5$, scale $s = \sigma \sqrt{(\nu - 2)/\nu}$ so that the variance is σ^2 .
 - ▶ Formulas: $\text{VaR}_\alpha = -(\mu + s q_\alpha)$ with $q_\alpha = t_\nu^{-1}(\alpha)$; $\text{ES}_\alpha = -\mu + s \frac{f_\nu(q_\alpha)}{\alpha} \frac{\nu + q_\alpha^2}{\nu - 1}$, with f_ν the Student- t density.
- ▣ **Tasks**
 - ▶ (a) Compute the scale s .
 - ▶ (b) Find $t_5^{-1}(0.01)$ and compute $\text{VaR}_{1\%}$ in % and in RON.
 - ▶ (c) Find $q = t_5^{-1}(0.025)$, compute the factors $f_5(q)/0.025$ and $(5 + q^2)/4$, then $\text{ES}_{2.5\%}$ in % and in RON.
 - ▶ (d) Compute the ratios Student- t /Normal for VaR and for ES (Normal values from A5), and the ratio ES/VaR.
- ▣ **Report**
 - ▶ Both measures in % and RON, and the three ratios.
- ▣ **Interpretation**
 - ▶ Which number changes more when the tails become heavy, VaR or ES, and what does this mean for capital?
- ▣ **Model:** A5

A8 [Proposed]: The Power of the Christoffersen Test — Task

Solution discussed in class

- ▣ **Question:** Can one year of breaches reveal that they cluster?
- ▣ **Data and assumptions**
 - ▶ Breaches I_t follow a two-state Markov chain with the correct unconditional rate $\pi = 1\%$ but clustered:
 $\pi_{11} = P(I_t = 1 \mid I_{t-1} = 1) = 0.20$ (independence would mean $\pi_{11} = 1\%$).
 - ▶ Christoffersen independence test $LR_{ind} \sim \chi^2(1)$, rejection at 5% if $LR_{ind} > 3.84$ [Christoffersen \(1998\)](#); a path without any breach counts as a non-rejection.
- ▣ **Tasks**
 - ▶ (a) Derive $\pi_{01} = P(I_t = 1 \mid I_{t-1} = 0)$ from the stationarity condition $\pi = \pi_{01}/(1 - \pi_{11} + \pi_{01})$.
 - ▶ (b) Simulate 20 000 paths of $T = 250$ days (first day a breach with probability π), compute LR_{ind} on each path and estimate the power as the share of rejections, with its Monte Carlo standard error.
 - ▶ (c) Repeat (b) for $T = 1\,000$ days with 5 000 paths.
 - ▶ (d) Compute the expected number of consecutive breach pairs $(T - 1)\pi\pi_{11}$ for both T .
- ▣ **Report**
 - ▶ π_{01} , both powers with the Monte Carlo standard error, the expected numbers of pairs.
- ▣ **Interpretation**
 - ▶ Why is independence almost untestable with one year of data?
- ▣ **Model:** A7

B6 [Proposed]: Backtesting VaR and ES for Bitcoin, 2019–2026 — Task

Solution discussed in class

- ▣ **Question:** Does the ranking of B5 hold for Bitcoin?
- ▣ **Data and assumptions**
 - ▶ Bitcoin daily losses (close in USD, course data), evaluation 1 Jan 2019 – 18 Sep 2026, $T = 2\,818$ days, every calendar day; each forecast uses only data up to $t - 1$.
 - ▶ The four models, windows and tests of B5; Newey–West variance with 8 lags; level 5% (10% for the model confidence set).
- ▣ **Tasks**
 - ▶ (a) Count the breaches of VaR 1% and compute the rate, p_{uc} and p_{ind} for each model.
 - ▶ (b) Run the DQ test with 4 lagged hits and VaR_t .
 - ▶ (c) Compute the FZ0 scores, the Diebold–Mariano tests and the model confidence set at 10%.
 - ▶ (d) Run the comparative ES backtest against HS.
 - ▶ (e) Plot the daily losses and the HS and FHS VaR 1% from 1 January 2022.
- ▣ **Report**
 - ▶ The two tables of B5 for Bitcoin and the chart of (e).
- ▣ **Interpretation**
 - ▶ Is the ranking of the models the same as for BET? Can the tests separate the models at all?
- ▣ **Model:** B5

A10 [Proposed]: Hurdles for One Hundred Factors — Task

Solution discussed in class

- ▣ **Question:** How high must the $|t|$ hurdle be when one hundred factors are tested?
- ▣ **Data and assumptions**
 - ▶ 100 candidate factors, each tested with a two-sided t -statistic that is approximately $N(0, 1)$ under H_0 ; target level 5%.
 - ▶ Bonferroni and BH as in A9; under arbitrary dependence the Benjamini–Yekutieli procedure replaces $0.05k/m$ by $0.05k/[m c(m)]$, $c(m) = \sum_{i=1}^m 1/i$ [Benjamini & Yekutieli \(2001\)](#).
- ▣ **Tasks**
 - ▶ (a) Compute the Bonferroni per-test level at FWER 5% and the hurdle $|t| > z_{1-0.05/(2m)}$.
 - ▶ (b) Compute the BH hurdle for $|t|$ when exactly 10 factors are declared and when only 1 is declared.
 - ▶ (c) Compute $c(100)$ and the Benjamini–Yekutieli hurdle when 10 factors are declared.
 - ▶ (d) Compare the hurdles with $|t| > 3$ proposed by [Harvey, Liu & Zhu \(2016\)](#).
- ▣ **Report**
 - ▶ The per-test levels and the four $|t|$ hurdles.
- ▣ **Interpretation**
 - ▶ Why does the BH hurdle depend on how many factors are declared, and is BH still valid when factors are positively correlated?
- ▣ **Model:** A9

A11 [Solved]: Why Kurtosis Fails with a Tail Index below 4 — Task

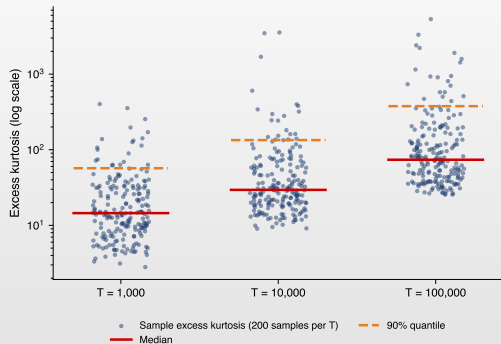
- ▣ **Question:** What does the sample kurtosis estimate when the tail index is below 4?
- ▣ **Data and assumptions**
 - ▶ $|X|$ has a Pareto-type tail: $P(|X| > x) = Cx^{-\alpha}$ for $x \geq x_0 > 0$, $C > 0$; α is the tail index.
 - ▶ Simulation design: Student- t with 3 degrees of freedom (tail index 3, finite variance), sample sizes $T = 10^3, 10^4, 10^5$, 200 samples of each size.
- ▣ **Tasks**
 - ▶ (a) Use $\mathbb{E}|X|^4 = \int_0^\infty 4x^3 P(|X| > x) dx$ to show that $\mathbb{E}|X|^4 = \infty$ when $\alpha \leq 4$.
 - ▶ (b) Simulate the samples, compute the sample excess kurtosis of each, and report its median and 90% quantile for each T .
 - ▶ (c) Explain what a bootstrap interval for the kurtosis estimates in this case [Athreya \(1987\)](#).
- ▣ **Report**
 - ▶ The proof of (a) and a table of the median and 90% quantile for the three sample sizes.
- ▣ **Interpretation**
 - ▶ What should you report instead of the kurtosis when the tail index is below 4?

A11 [Solved]: Solution (1/2)

- (a) Step 1: for $x \geq x_0$ the integrand is $4x^3 \cdot Cx^{-\alpha} = 4Cx^{3-\alpha}$, and it is non-negative everywhere.
- Step 2: $\mathbb{E}|X|^4 \geq 4C \int_{x_0}^{\infty} x^{3-\alpha} dx$, and this integral diverges when $3 - \alpha \geq -1$, i.e. $\alpha \leq 4$.
- Step 3: for $2 < \alpha \leq 4$ the variance is finite but $\mathbb{E}X^4 = \infty$, so the population kurtosis $\mathbb{E}X^4/(\mathbb{E}X^2)^2$ is infinite and the sample kurtosis has no finite limit.
- (b) Steps: draw 200 samples of each size, compute the sample excess kurtosis, take the median and the 90% quantile across the 200 values.

T	10^3	10^4	10^5
Median excess kurtosis	14.5	29.4	73.7
90% quantile	56.9	134.0	376.4

A11 [Solved]: Solution (2/2)



- Reading: the median grows with T and the 90% quantile grows faster: the statistic does not settle down, it grows without limit.
- (c) There is no population value for an interval to cover, and the bootstrap of such moments is inconsistent [Athreya \(1987\)](#): a bootstrap interval for the kurtosis has no coverage guarantee.
- Interpretation: when the tail index may be below 4, report a tail index with its interval (Hill with block bootstrap, B1) and not the kurtosis; a large sample kurtosis says only that the tails are heavy, not how heavy.

B2 [Proposed]: Stylised Facts of Bitcoin — Task

Solution discussed in class

- **Question:** Do the stylised facts of B1 hold for Bitcoin?
- **Data and assumptions**
 - ▶ Bitcoin in USD (daily close, course data), every calendar day, 18 Sep 2014 – 18 Sep 2026: $T = 4\,384$ log returns in %.
 - ▶ Same settings as B1: $k = 5\%$ of the loss days, 999 block-bootstrap resamples with 20-day blocks, 10 lags, level 5%; BET results from B1.
- **Tasks**
 - ▶ (a) Compute the standard deviation, skewness and excess kurtosis.
 - ▶ (b) Estimate the Hill tail index with the asymptotic and the block-bootstrap 95% CI.
 - ▶ (c) Compute $Q(10)$ on r_t and r_t^2 and the robust $Q^*(10)$, and decide at 5%.
 - ▶ (d) Count the days beyond ± 4 standard deviations and compare with the Normal distribution.
 - ▶ (e) Test $H_0: \alpha_{\text{BTC}} = \alpha_{\text{BET}}$ with $z = (\hat{\alpha}_{\text{BTC}} - \hat{\alpha}_{\text{BET}}) / \sqrt{\text{se}_{\text{BTC}}^2 + \text{se}_{\text{BET}}^2}$ using the block-bootstrap standard errors (A12).
- **Report**
 - ▶ A table Bitcoin vs BET with (a)–(d); the test of (e) with z , p and decision.
- **Interpretation**
 - ▶ Are Bitcoin's tails heavier than the BET's, or is Bitcoin only more volatile?
 - ▶ Does the linear autocorrelation of Bitcoin survive the robust test?
- **Model:** B1

A12 [Proposed]: Comparing Two Tail Indices — Task

Solution discussed in class

- ▣ **Question:** Are the tails of Bitcoin heavier than those of the BET?
- ▣ **Data and assumptions**
 - ▶ Hill estimator on the k largest losses: $\hat{\alpha}^{-1} = \frac{1}{k} \sum_{i=1}^k \ln L_{(i)} - \ln L_{(k+1)}$, $k = 5\%$ of the loss days; $\text{se}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$ for i.i.d. data [Hill \(1975\)](#).
 - ▶ From B1–B2: BET $\hat{\alpha} = 2.39$ ($k = 153$), Bitcoin $\hat{\alpha} = 2.74$ ($k = 104$); block-bootstrap standard errors (999 resamples, 20-day blocks): BET 0.233, Bitcoin 0.227; simplifying assumption: the two estimates are independent (the samples overlap in time, so common shocks would add $-2 \text{Cov}(\hat{\alpha}_{\text{BTC}}, \hat{\alpha}_{\text{BET}})$ to the variance of the difference); level 5%.
- ▣ **Tasks**
 - ▶ (a) Compute the i.i.d. standard errors of both estimates.
 - ▶ (b) Test $H_0: \alpha_{\text{BTC}} = \alpha_{\text{BET}}$ with $z = (\hat{\alpha}_{\text{BTC}} - \hat{\alpha}_{\text{BET}}) / \sqrt{\text{se}_{\text{BTC}}^2 + \text{se}_{\text{BET}}^2}$, two-sided, with the i.i.d. standard errors.
 - ▶ (c) Repeat the test with the block-bootstrap standard errors.
- ▣ **Report**
 - ▶ All standard errors, both z-statistics, p -values and decisions.
- ▣ **Interpretation**
 - ▶ Why can the i.i.d. formula misstate the uncertainty for returns, and does the conclusion depend on it?
- ▣ **Model:** A11, B1

B8 [Proposed]: The Calendar Trap: Bitcoin and SPY — Task

Solution discussed in class

- ▣ **Question:** What does aligning returns instead of prices do to Bitcoin–SPY statistics?
- ▣ **Data and assumptions**
 - ▶ Bitcoin (daily close in USD, every calendar day) and SPY, the S&P 500 ETF (close adjusted for dividends and splits, weekdays only), from the course data; daily log returns in %.
 - ▶ Two periods: 1 January 2015 – 31 December 2019 (1 258 common days) and 1 January 2020 – 18 September 2026 (1 687 common days).
- ▣ **Tasks**
 - ▶ (a) Wrong alignment: compute the log returns of each asset on its own calendar, then keep the common days.
 - ▶ (b) Right alignment: keep the common days of the prices, then compute the log returns (a Monday return then covers Friday to Monday for both assets).
 - ▶ (c) For each period and each alignment compute the Bitcoin–SPY correlation, the standard deviation of Bitcoin returns on all days and on Mondays, and the cumulative Bitcoin return in %, $100[\exp(\sum_t r_t/100) - 1]$ (with r_t in %).
 - ▶ (d) Plot the 250-day rolling correlation from 2016 for both alignments.
- ▣ **Report**
 - ▶ A table (statistics $\times$ periods $\times$ alignments) and the chart of (d).
- ▣ **Interpretation**
 - ▶ Which statistic is most damaged by the wrong alignment? Would the VaR of a long-only Bitcoin–SPY portfolio, from a variance–covariance model built on it, be too low or too high?
- ▣ **Model:** B7

C3 [Proposed]: Nonstandard Errors: A Mini-Multiverse — Task (1/2)

Solution discussed in class

▣ **Question:** How much do the analysis choices alone move a market-efficiency trend?

▣ **Data and assumptions**

- ▶ Context: in [Menkveld et al. \(2024\)](#) 164 teams tested the same hypotheses on the same data; the IQR of their yearly market-efficiency trend estimates (the nonstandard error) was 6.7 pp, against a median SE of 2.5% (Table I, Panel C).
- ▶ Data: daily closes of the Euro Stoxx 50 index and of the BET (course data), 2002–2018, then up to the last full year; H_0 : “market efficiency has not changed over time” (Sec. I.A).

▣ **Tasks**

- ▶ (a) Before any run, commit a plan with the forks of Table V that daily data allow: outliers (none; annual m_t winsorised or trimmed at its 2.5 and 97.5 percentiles, trimmed years left missing); measure, within each year ($|VR(q) - 1|$, $VR(q) = \text{Var}(\sum_{i=0}^{q-1} r_{t-i}) / (q \text{Var}(r_t))$), overlapping sums, $q = 5, 21$; R^2 of an AR(1) with intercept); model (linear trend; log difference; relative difference).
- ▶ (b) For each index and each of the 27 paths (3 measures $\times$ 3 outlier rules $\times$ 3 models), compute the annual measure m_t and its average yearly % change with an SE: $100 b/\bar{m}$ from the regression of the measure on a linear trend, the mean of $100 \Delta \ln m_t$, or the mean of $100 (m_t/m_{t-1} - 1)$. Nulls: for a trendless stationary m_t the slope and mean log change are centred at zero, the mean relative change is not.
- ▶ (c) Report the median, the IQR (the nonstandard error of your multiverse) and the 10–90 percentile range of the 27 estimates; compare the IQR with $1.35 \times$ the median SE.

C3 [Proposed]: Nonstandard Errors: A Mini-Multiverse — Task (2/2)

Solution discussed in class

▣ Tasks (continued)

- ▶ (a, continued) The plan also fixes the rule for zero or missing m_t , the Holm family (27 tests per index and period) and the last complete year at your data cutoff.
- ▶ (d) Rank the forks by the k -sample Anderson–Darling statistic (`scipy.stats.anderson_ksamp`) of the estimates grouped by the alternative at each fork (Sec. II.C).
- ▶ (e) Test each path's trend against zero at 5% with the Holm correction over the 27 tests, and report how many paths reject with each sign (positive trend = declining efficiency).

▣ Report

- ▶ A table of the 27 estimates with their SEs for each index; the median, IQR and 10–90 range; the Anderson–Darling ranking of the forks; the number of Holm rejections by sign; a box plot of the estimates grouped by the most influential fork.
- ▶ Model: A9 (Holm and BH over a family of tests); the lecture case study

▣ Interpretation

- ▶ Is the IQR over 27 paths an estimate of the nonstandard error of 164 human teams? Use footnote 32 of the paper in your answer.
- ▶ If the sign of the trend changes with the model fork, what should your project report, and which rule would you have written into the pre-analysis plan?

References (1/3)

- ▣ Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- ▣ Andrews, D.W.K. (1999). Estimation when a parameter is on a boundary. *Econometrica*, 67(6), 1341–1383.
- ▣ Athreya, K.B. (1987). Bootstrap of the mean in the infinite variance case. *Annals of Statistics*, 15(2), 724–731.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *Annals of Statistics*, 29(4), 1165–1188.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.
- ▣ Christoffersen, P.F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862.
- ▣ Davis, R.A., Mikosch, T. (1998). The sample autocorrelations of heavy-tailed processes with applications to ARCH. *Annals of Statistics*, 26(5), 2049–2080.
- ▣ Engle, R.F., Manganelli, S. (2004). CAViaR: conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.

References (2/3)

- ▣ Harvey, C.R., Liu, Y., Zhu, H. (2016). ... and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.
- ▣ Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- ▣ Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- ▣ Künsch, H.R. (1989). The jackknife and the bootstrap for general stationary observations. *Annals of Statistics*, 17(3), 1217–1241.
- ▣ Kupiec, P.H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84.
- ▣ Lobato, I., Nankervis, J.C., Savin, N.E. (2001). Testing for autocorrelation using a modified Box–Pierce Q test. *International Economic Review*, 42(1), 187–205.
- ▣ McNeil, A.J., Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press.
- ▣ Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). Nonstandard errors. *Journal of Finance*, 79(3), 2339–2390.
- ▣ Nelson, D.B. (1990). Stationarity and persistence in the GARCH(1,1) model. *Econometric Theory*, 6(3), 318–334.
- ▣ Nolde, N., Ziegel, J.F. (2017). Elicitability and backtesting: perspectives for banking regulation. *Annals of Applied Statistics*, 11(4), 1833–1874.

References (3/3)

- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#). *European Journal of Finance*, 29(10), 1064–1105.